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Nuclear magnetic resonance investigations of aluminum containing phosphate glass-ceramics

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Abstract

Glasses and glass-ceramics of the system P_2O_5 – CaO – Na_2O – $\{Al_2O_3\}$ have been studied by ^{31}P and ^{27}Al and two-dimensional correlation nuclear magnetic resonance. The basic glasses have a chain-like structure including amorphous diphosphate and longer chains having at least four units. The Q^1/Q^2 group ratios depend on both the content of network-modifying oxides and aluminum oxide. Crystallization of the glasses involves the conversion of glassy metaphosphate sites into crystalline pyro- and orthophosphate sites. ^{27}Al correlation experiments do not detect the presence of Al – O – Al bonds in the glasses and in the glass-ceramics. © 2000 Elsevier Science B.V. All rights reserved.

1. Introduction

Phosphate glasses and glass-ceramics in the diphosphate/monophosphate region are interesting in view of bioactive longtime stable and resorbable implant materials [1,2]. The development of glass-ceramic materials possessing well-defined biological properties is facilitated by knowledge of the glass structure. The effect of aluminum oxide contaminations (caused by crucible dissolution) on the structure and the crystallization of the phosphate glasses is known [3–5]. However, the structural role of alumina in sodium calcium phosphate glasses is not completely explained. The aim of the present work is to analyze the effect of aluminum on the glass structure and the crystallization of the glasses in the system P_2O_5 – CaO – Na_2O – $\{Al_2O_3\}$. Especially, the destruction of phosphate chains

and rings during crystallization of such glasses are investigated. Two typical base glasses (GI and GII) were modified by adding aluminum oxide up to 6 mol%.

Nuclear magnetic resonance (NMR) could be used to determine the distribution of phosphorus and aluminum over the various coordination polyhedra. Furthermore, 2D NMR yields direct information about the medium range order, especially about the connectivity of the coordination polyhedra (e.g., $Q^{[n]}$ units, PO_4 -tetrahedra with n bridging oxygen atoms) [6–8]. With newly developed experimental techniques (off-angle spinning method [9]) the connections of AlO_x polyhedra were detected.

2. Experimental

The glasses were prepared by melting calcium metaphosphate, sodium carbonate, sodium metaphosphate and aluminum oxide in SiO_2 crucibles

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Table 1

Compositions of the glasses in mol% (synthesis) and T_g obtained by DTA

Glass	P ₂ O ₅ (mol%)	CaO (mol%)	Na ₂ O (mol%)	Al ₂ O ₃ (mol%)	T_g (°C)
GI	39.4	33.4	27.2	0	375 ± 5
GI/1	38.2	32.4	26.4	3	427 ± 5
GI/2	37	31.4	25.6	6	438 ± 5
GII	43.6	30.6	25.8	0	363 ± 5
GII/1	42.7	30	25.3	2	390 ± 5
GII/2	41.9	29.4	24.7	4	422 ± 5

in an electrically-heated furnace. The melts were poured into water and remelted in platinum crucibles at 1300°C for 1 h. Their synthesis compositions are listed in Table 1 together with their transformation temperatures (T_g) which change with the aluminum content as determined by difference thermal analysis (DTA) (Shimadzu DTA 50). To avoid fictive temperature effects caused by different thermal histories, all samples were produced and measured under same conditions (identical melting temperatures, melting times, cooling rates and using the same melting techniques). The crystallization of the glasses was induced by a heat treatment in the range of 375–700°C. A glass-ceramic sample, GI/2, was remelted to determine if the structural changes during heat treatment are reversible. The melting procedure was the same as described above. The NMR spectra were acquired (on a Bruker AMX 400 spectrometer) with a magic angle spinning (MAS) frequency $\nu_r < 15$ kHz. The ^{31}P chemical shift tensor parameters were obtained from slow spinning MAS spectra ($\omega_R/2\pi = 5$ kHz) using the method

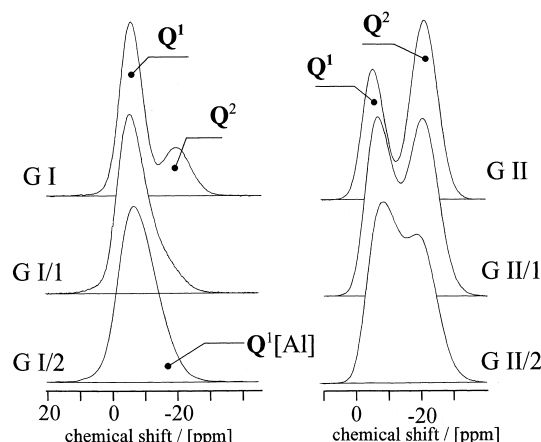


Fig. 1. ^{31}P MAS NMR spectra of the glasses (resonance frequency $\nu_0 = 161.92$ MHz, spinning frequency $\omega_R/2\pi = 14.0$ kHz, pulse width $t_w = 2$ μs , and repetition time $t_r = 200$ s).

of Herzfeld and Berger [10]. The chemical shifts are referenced to a solution of phosphoric acid (85%) for ^{31}P and to a solution of AlCl_3 (1 M) for ^{27}Al NMR.

3. Results

Fig. 1 shows the ^{31}P MAS NMR spectra of the glasses (central lines only). The fitted chemical shift tensor parameters and the relative Q^n group content of the samples are listed in Table 2. For analyzing the intermediate range phosphate structure, 2D double quantum (DQ) NMR spectra have been acquired. As an example, Fig. 2 shows the 2D DQ spectrum of the sample GII/1. It

Table 2

Chemical shift tensor parameter of the Q^1 and Q^2 groups and relative fraction of the different Q^n groups as a function of the aluminum content of the glasses^a

Glass	Q^1				Q^2			
	δ_{iso} (ppm)	$\Delta\delta$ (ppm)	η	Q^1 (%)	δ_{iso} (ppm)	$\Delta\delta$ (ppm)	η	Q^2 (%)
GI	-5.7 ± 0.3	139 ± 5	0.44 ± 0.1	68 ± 3	-19.4 ± 0.3	-177 ± 5	0.65 ± 0.1	32 ± 2
GI/1	-6.3 ± 0.3	132 ± 5	0.68 ± 0.1	87 ± 5	-17.1 ± 0.3	-148 ± 5	0.55 ± 0.1	13 ± 1
GI/2	-6.9 ± 0.3	124 ± 5	0.68 ± 0.1	76 ± 5				
	-14.5 ± 0.5	138 ± 5	0.49 ± 0.1	24 ± 2				
GII	-5.9 ± 0.3	142 ± 5	0.74 ± 0.1	33 ± 2	-21.7 ± 0.3	-203 ± 5	0.46 ± 0.1	67 ± 3
GII/1	-7.4 ± 0.3	138 ± 5	0.74 ± 0.1	43 ± 2	-20.8 ± 0.3	-191 ± 5	0.48 ± 0.1	57 ± 3
GII/2	-8.4 ± 0.3	135 ± 5	0.83 ± 0.1	46 ± 2	-19.6 ± 0.3	-178 ± 5	0.44 ± 0.1	54 ± 3

^a The parameters were obtained by line fit analysis of slow spinning spectra.

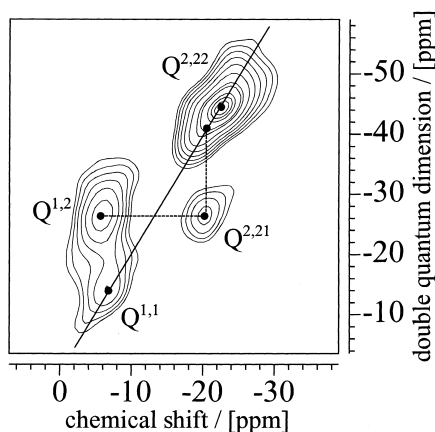


Fig. 2. 2D double quantum correlation spectrum [8,12] of the basic glass GII. The resonance frequency was $\nu_0 = 161.92$ MHz, spinning frequency $\omega_R/2\pi = 12.5$ kHz, pulse width $t_w = 5$ μ s, and repetition time $t_r = 200$ s. The double quantum excitation and reconversion periods were each 640 μ s. The 256×64 data points were collect rotor-synchronized, in States method, and with 256 scans per point.

consists of DQ peaks at $f_2/f_1 = -6.3/-12.6$ ppm attributed to $Q^{1,1}$ groups (Q^1 groups connected to a second Q^1 group), at $f_2/f_1 = -5.7/-26.2$ ppm attributed to $Q^{1,2}$ groups (Q^1 groups connected to a Q^2 group), and the corresponding $Q^{2,21}$ groups (Q^2 groups connected to a Q^1 group and a Q^2 group) at $f_2/f_1 = -20.0/-26.2$ ppm. Furthermore, $Q^{2,22}$ group peaks (Q^2 groups connected to other Q^2 groups) occur at $f_2/f_1 = -22.5$ ppm/–45 ppm. The ^{31}P MAS NMR spectra of the heat-treated samples GII/1 are shown in Fig. 3 as a function of the crystallization temperature. The isotropic shifts of the various lines of the crystalline phases are listed in Table 3. In Fig. 4 the 2D RFDR correlation spectrum [11] of the glass-ceramic GII/1 and few of the cross-sections are plotted. Besides the Q^2 exchange peaks (lines a, b; see Table 3), the main part of the spectrum (lines c, d, e, f, h; see Table 3) consists of exchange peaks which we attribute to a mixed phase of calcium sodium aluminum diphosphates, of the peaks of a crystalline calcium diphosphate (line g) and a crystalline sodium diphosphate (line i, j). In Fig. 5 the integrated Q^2 group content of the samples GII as a function of the crystallization temperature minus transformation temperature (T_g).

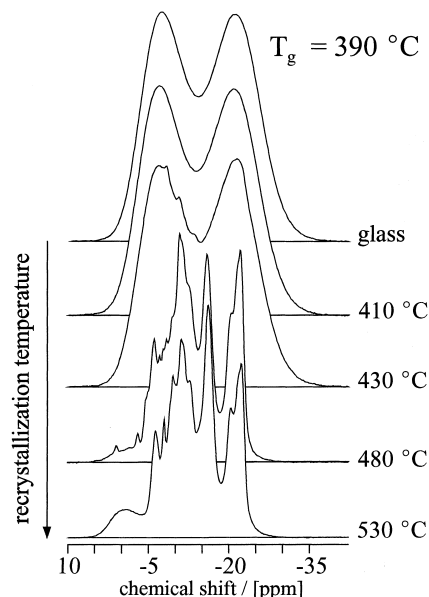


Fig. 3. ^{31}P MAS NMR spectra of the crystallized glass GII/1 as a function of the crystallization temperature (for 24 h). Resonance frequency $\nu_0 = 161.92$ MHz, spinning frequency $\omega_R/2\pi = 14.0$ kHz, pulse width $t_w = 2$ μ s and repetition time $t_r = 200$ s.

Table 3

Typical isotropic chemical shifts of the various Q^1 and Q^2 line positions in the crystallized glasses^a

Line	δ_{iso} (ppm)	Structural units
a	-22.0 ± 0.3	$Q^2[\text{Na}]$
b	-20.2 ± 0.3	$Q^2[\text{Na}]$
c	-16.1 ± 0.3	$Q^1[\text{Na,Ca,Al}]$
d	-13.1 ± 0.3	$Q^1[\text{Na,Ca,Al}]$
e	-11.4 ± 0.3	$Q^1[\text{Na,Ca,Al}]$
f	-9.6 ± 0.3	$Q^1[\text{Na,Ca,Al}]$
g	-7.5 ± 0.3	$Q^1[\text{Ca}]$
h	-6.0 ± 0.3	$Q^1[\text{Na,Ca,Al}]$
i	-4.0 ± 0.5	$Q^1[\text{Na}]$
j	1.5 ± 0.5	$Q^1[\text{Na}]$

^a Not all samples possess each line positions listed here. The line positions of amorphous phases (e.g., glass matrix) are not included.

Fig. 6 shows the ^{27}Al MAS spectra of the sample GI/2, the heat-treated sample GI/2 (550°C, 24 h), and the remelted sample. The spectrum of the sample contains three lines attributed to fourfold ($\delta_{\text{iso}} = 43 \pm 10$ ppm), fivefold ($\delta_{\text{iso}} = 13 \pm 5$ ppm), sixfold ($\delta_{\text{iso}} = -13 \pm 5$ ppm) [4] oxygen-

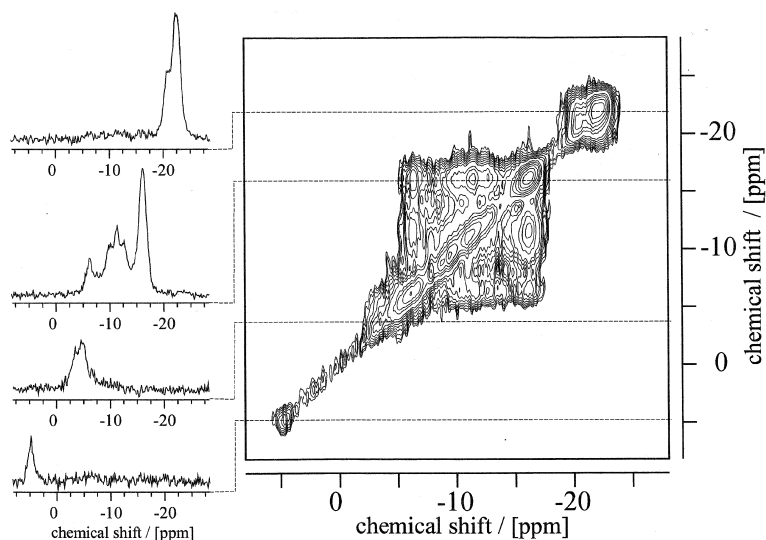


Fig. 4. 2D RFDR correlation spectrum [11] of the glass ceramic GII/1 (530°C, 24 h). The displayed cross-sections of the 2D spectrum show the different phases in the ceramic. The resonance frequency was $\nu_0 = 161.92$ MHz, spinning frequency $\omega_R/2\pi = 12.5$ kHz, pulse width $t_w = 2$ μ s, $t_{\text{mix}} = 25$ ms and repetition time $t_r = 200$ s. The 256×128 data points were collect rotor-synchronized, in TPPI method, and with 32 scans per point.

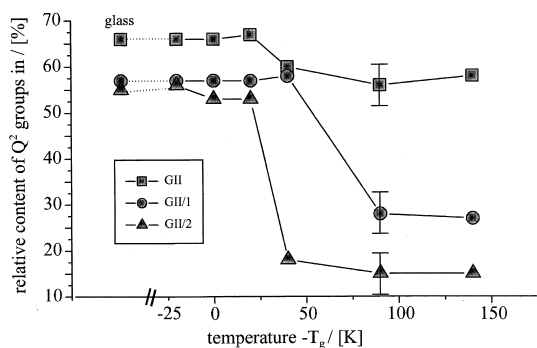


Fig. 5. Relative content of Q^2 groups in the glasses GII as a function of the difference between crystallization temperature and transformation temperature (T_g). The Q^2 fractions were obtained from line fit analysis of the MAS spectra, integrated of all corresponding line positions ($T_g(\text{GII}) = 363^\circ\text{C}$, $T_g(\text{GII}/1) = 391^\circ\text{C}$, and $T_g(\text{GII}/2) = 422^\circ\text{C}$).

coordinated aluminum with phosphorus in the second coordination sphere. The aluminum distribution over the various coordination numbers in the glasses as a function of the crystallization temperature is summarized in Table 4. The crystallized glasses (except for heat treatment below T_g) have only four- and sixfold coordinated alu-

minum. For verifying that aluminum is coordinated by phosphorus in the second coordination sphere and not by other aluminum ions, 2D off-angle spinning has been applied [9]. Fig. 7 shows the off-angle spectrum of the sample GI/2. No cross-peaks between the various coordination states are observed.

4. Discussion

Irrespective of the contents of network modifying oxides and aluminum oxide the ^{31}P MAS NMR spectra of all samples (except for sample GI/2) have two NMR lines attributed to Q^1 and Q^2 groups. In sample GI/2 a second type of Q^1 group occurs at $\delta_{\text{iso}} = -14.5$ ppm. These Q^1 can be attributed to amorphous diphosphates affected by sodium and aluminum because crystalline-mixed aluminum sodium diphosphates have similar chemical shifts. The basic glass GI possesses a Q^1/Q^2 ratio of 2.1. The smaller network-modifying oxide content of the sample GII results in an increased fraction of Q^2 groups ($Q^1/Q^2 = 0.49$). The fraction of Q^2 groups of the aluminum containing

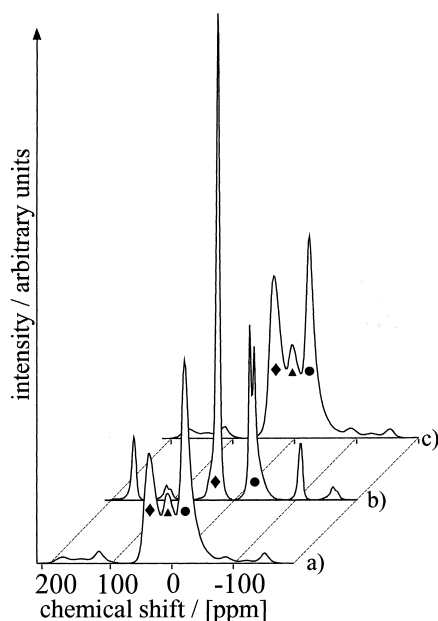


Fig. 6. ^{27}Al MAS NMR spectra of the glass GI/2, the crystallized glass GI/2 (550°C, 24 h), and the remelted sample. The line positions of fourfold oxygen-coordinated aluminum are marked by (◆), the fivefold by (▲) and, the sixfold by (●) (resonance frequency $\nu_0 = 104.7$ MHz, spinning frequency $\omega_R/2\pi = 15.0$ kHz, pulse width $t_w = 1$ μs and repetition time $t_r = 2$ s).

samples decreases proportionally to the content of aluminum, while the fraction of the Q^1 groups increases. Additionally, the shift tensor parameters of Q^1 and Q^2 groups change monotonically with the aluminum content of the glasses (as seen in Table 2). The decreasing chemical shifts indicate the increasing affect of the charged aluminum on the phosphate anion. Obviously, the aluminum acts as a network-modifying oxide. The intermediate structure of the glasses could be described by the extended $Q^{n,kl}$ group nomenclature [12]. In the DQ spectrum (Fig. 2), the $Q^{1,1}$ peaks are caused by amorphous diphosphate like anion structures in the glasses. The $Q^{1,2}$ and $Q^{2,21}$ peaks can be attributed to chain-like structures. The intensive $Q^{2,22}$ DQ peaks prove the occurrence of long phosphate chains or rings.

The ^{31}P NMR spectra of the heat-treated samples differ with respect to their line positions and line-widths. Glasses heat-treated at $T_g + 40$ K or less, (24 h) possess the typical glass spectrum. Heat treating at higher temperatures, however, results in a complete crystallization of the samples. From the 2D RFDR spectrum (Fig. 4 for sample GII/1) an unambiguous assignment of the line positions caused by different phases in the glass-ceramic can be made. The phase distribution of the glass-ceramic GII/1 is typical for the investigated glass-

Table 4

Relative content of the different aluminum coordination in selected glasses and crystallized glasses obtained by line fit analysis of the ^{27}Al MAS NMR spectra^a

	Al[4]-O (%)	Al[5]-O (%)	Al[6]-O (%)
GI/1 glass	8.0 ± 1	10.7 ± 1	81.3 ± 5
GI/1 525°C	0	0	100 – 5
GI/1 550°C	0	0	100 – 5
GI/2 glass	28.6 ± 3	24.2 ± 2	47.2 ± 5
GI/2 475°C	36.8 ± 4	22.2 ± 2	41.0 ± 4
GI/2 500°C	73.4 ± 5	0	26.6 ± 3
GI/2 525°C	58.2 ± 5	0	41.8 ± 4
GI/2 550°C	62.5 ± 5	0	37.5 ± 4
GI/2 remelted glass	31.2 ± 3	26.5 ± 2	42.4 ± 4
GI/2 remelted glass	3.4 ± 1	3.0 ± 1	90.6 ± 5
GII/1 glass	0	0	100 – 5
GII/1 480°C	0	0	100 – 5
GII/1 530°C	0	0	100 – 5
GII/2 glass	5.0 ± 1	8.3 ± 1	86.7 ± 5
GII/2 512°C	0	0	100 – 5
GII/2 562°C	0	0	100 – 5

^a The notation Al^[n] indicates the n -fold coordination of the Al site.

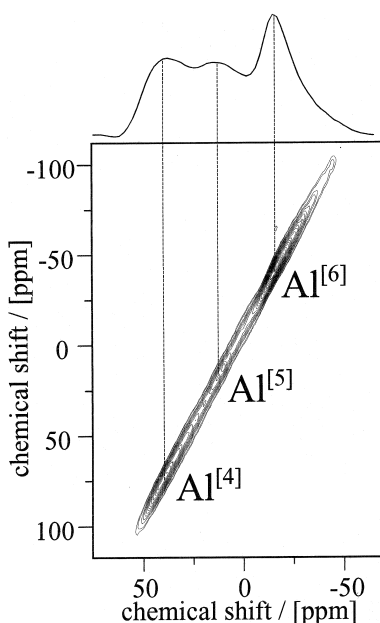


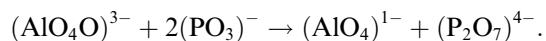
Fig. 7. ^{27}Al off-angle spectrum [9] of the glass GI/2. The resonance frequency was $\nu_0 = 78.5$ MHz, the pulse width $t_w = 2.9$ μs and the repetition time $t_r = 1$ s. During mixing ($t_{\text{mix}} = 50$ ms) the sample was spinned at $\Theta_{\text{off}} = 50^\circ$ with a spinning frequency of $\omega_R/2\pi = 10.0$ kHz. The 256×256 data points were collected rotor-synchronized, in TPPI method, and with 128 scans per point.

ceramics (except for GI/2). The Q^2 groups (line a, b; see Table 3) possess a cross-peak which we suggest is due to their proximity. Q^2 – Q^1 cross-peaks are not observed. The Q^2 groups are therefore attributed to phosphate chains or phosphate rings. The main part of the spectrum consists of exchange peaks between the lines c, d, e, f and h (see Table 3). We suggest that the complex exchange peak pattern shows that these Q^1 groups belong to a single mixed Ca, Na, Al diphosphate phase. Simultaneously, aluminum is fourfold and sixfold oxygen-coordinated. In spite of different aluminum contents, the structures of the mixed Ca, Na, Al diphosphate phases are similar as indicated by nearly unchanged isotropic chemical shifts in the ^{31}P and ^{27}Al spectra. X-ray investigations on a mixed calcium aluminum and sodium aluminum diphosphate [13] confirm the variety of mixed aluminum diphosphate structures.

The Q^1/Q^2 ratio of the crystallized glasses as a function of the crystallization temperature is most

interesting to us (Fig. 5). Irrespective of the aluminum content, the Q^2 fraction of the glasses is stable until $T_g + 25$ K. Crystallizing at higher temperatures results in a decondensation of the Q^2 groups. The relative Q^2 loss depends on the aluminum content of the glasses. Whereas in the aluminum free sample, GII, a Q^2 loss of only 8% is found, the numbers are 30% and 40% for the samples GII/1 and GII/2, respectively. We note that sample GI/2 contains only Q^1 groups. Crystallizing that glass results in a decomposition of the Q^1 groups to Q^0 units. It is obvious that the decomposition of the phosphate structure upon crystallizing is related to the aluminum content of the glasses. At the same time, the distribution of aluminum over the various coordination numbers changes as well with greater fractions in the sixfold coordination state with higher temperatures and aluminum contents of the base glasses (Table 4). Possibly, the decreasing content of fivefold aluminum (only present in the base glasses and sample GI/2 tempered at 475°C) during heat treating is accompanied by the formation of the mixed crystalline aluminum calcium sodium diphosphates. Furthermore, the off-angle correlation spectrum (Fig. 7) shows that in these glasses, the aluminum is coordinated by $[\text{O}-\text{P}]_n$ exclusively. That result is consistent with the assumption that aluminum acts as a network-modifying oxide in such glasses. Finally, the NMR spectra of the remelted glass-ceramics are identical to that of the original glasses. Therefore, melting relics and effects of melting balance can be excluded.

Summarizing, for both the GI and GII series, glassy Q^2 groups are converted to crystallized Q^1 (or Q^1 to Q^0) tetrahedra. Additionally, $\text{Al}[5]$ sites (the symbol $[n]$ means n -fold coordinated) in series GI are replaced by crystalline $\text{Al}[4]$ -sites, whereas $\text{Al}[4]$ and $\text{Al}[5]$ -sites in series GII are replaced by crystalline $\text{Al}[6]$ -sites. The crystallization process in series GI can be understood by the following reaction:



The situation is more complicated in series GII since it appears that the phosphorus and aluminum anions have more negative valences in the glass-ceramics than in the glasses. Possibly,

atmospheric oxygen has an important role in the crystallization process. The NMR spectra of vacuum-crystallized glasses GII and ‘normal’ crystallized samples GII are quite different. However, the investigations are in process and the mechanism is not understood.

5. Conclusions

The investigated glasses of the system P_2O_5 – CaO – Na_2O – $\{\text{Al}_2\text{O}_3\}$ possess a chain like phosphate structure. The Q^1/Q^2 ratios depend on both the content of network-modifying oxides and aluminum. Increasing contents of CaO , Na_2O and aluminum oxide result in decreasing fractions of Q^2 groups. Apparently, aluminum acts as a network-modifying oxide in these types of glasses. The base glasses possess a chain-like structure including amorphous diphosphate, and longer chains having at least four units. Crystallization of the glasses at temperatures $T_g + 25$ K results in formation of different crystalline phosphates and decrease of the averaged condensation degree of the phosphate anion structure. The fraction of Q^1 (as well as Q^0 groups) increases with the crystallization temperature (relative to T_g) while the fraction of Q^2 groups decreases. Simultaneously, the aluminum coordination changes. The typical fivefold oxygen-coordinated aluminum of the glasses vanishes in the glass-ceramics. Obviously, the phosphate chain (or ring) destruction and the change of aluminum coordination during heat treatment are associated with formation of a stable crystalline aluminum calcium sodium diphosphate phase in the glass-ceramics. However, the scission of P–O–P bonds

requires additional oxygen to saturate the newly formed free phosphorus valences. Because of the relative low content of the fivefold oxygen coordinated aluminum, the presence of these species alone cannot explain the observed strong decondensation of the phosphate anions.

Acknowledgements

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